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Inventor(s)	Jordan; David Scott et al.

Pipe fittings

Abstract

Stab-type pipe fittings capable of being inserted onto plastic pipes are provided. The pipe fittings have an outer casing with pipe ports, a hollow inner casing positioned within the outer casing, a pipe gripper and stiffener positioned within the inner casing. The pipe ports have an elongated lip that straightens a pipe inserted into the fitting. The stiffener provides a multi-compression seal to facilitate the leak proof connection.

Inventors: Jordan; David Scott (Tulsa, OK), Wartluft; Donald Wayne (Tulsa, OK)

Applicant: Hubbell Incorporated (Shelton, CT)

Family ID: 60940491

Assignee: Hubbell Incorporated (Shelton, CT)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending application Ser. No. 15/644,549 filed on Jul. 7, 2017 (now U.S. Pat. No. 11,060,647) and claims benefit from U.S. Provisional Application Ser. No. 62/361,299 filed on Jul. 12, 2016 the entire contents of both are herein incorporated by reference.

BACKGROUND

Field

(1) The present disclosure relates to stab-type pipe fittings designed to couple one or more plastic pipes to a fitting, and more particularly to stab-type pipe fittings with an improved seal between the fitting and the pipe.

Description of the Related Art

(2) Stab-type pipe fittings are designed to be inserted manually onto the end of a length of plastic pipe or between lengths of plastic pipes. Generally, due to mechanisms within stab-type fittings, after a fitting is inserted onto a length of plastic pipe the fitting is not intended to be pulled off of the pipe, such that the way to remove the fitting is to cut the pipe. Thus, stab-type pipe fittings are a one-time use pipe fitting that facilitates quick and easy connection to plastic pipes. If properly installed, stab-type pipe fittings form a sealed, leak proof pathway between the plastic pipes inserted into the fittings.

SUMMARY

(3) The present disclosure provides exemplary embodiments of improvements to stab-type pipe fittings that are capable of being inserted onto plastic pipes and provide a leak proof connection. In one exemplary embodiment, the pipe fitting comprises an outer casing having at least one pipe port providing a passageway for a pipe to be passed into the outer casing, a hollow inner casing positioned within the outer casing and having at least one opening aligned with the at least one pipe port, at least one gripper positioned within the inner casing, and a stiffener positioned at least partially within the inner casing and the at least one gripper opening. The at least one pipe port comprises an elongated lip and a port opening forming at least part of the passageway.

(4) The at least one gripper is positioned within the inner casing and has an opening aligned with the at least one pipe port. The at least one gripper includes one or more reverse serrations that can flex when a pipe is passed through the at least one gripper opening imparting little resistance to the forward advancement of the pipe into the at least one gripper while imparting sufficient resistance to rearward movement of the pipe to prevent withdrawal of the pipe from the at least one gripper.

(5) The stiffener has at least a pair of sealing members each positioned within a groove in an outer surface of the stiffener such that when a pipe is passed over the stiffener and sealing members the sealing members can contact an interior wall of the pipe to provide a seal between the pipe and the stiffener. In one embodiment, the pair of sealing members have the same diameter and the grooves have different dimensions. In another embodiment, the pair of sealing members have different diameters and the grooves have the same dimensions.

(6) In another embodiment, the stiffener has at least three sealing members each positioned within a groove in an outer surface of the stiffener such that when a pipe is passed over the stiffener and sealing members the sealing members can contact an interior wall of the pipe to provide a seal between the pipe and the stiffener. In one embodiment, the at least three sealing members have the same diameter and the grooves have different dimensions. In another embodiment, the at least three sealing members have different diameters and the grooves have the same dimensions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete appreciation of the present disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective view of an exemplary embodiment of a pipe fitting according to the present disclosure;
- (3) FIG. 2 is a cross-sectional view of the pipe fitting of FIG. 1 taken from line 2-2 and illustrating a pipe being inserted into the fitting;
- (4) FIG. 3 is a perspective view of the pipe fitting of FIG. 1 with a portion of an outer casing removed to reveal an inner casing encased with the outer casing;
- (5) FIG. 4 is an exploded view of the inner casing of FIG. 3 revealing a pair of grippers separated by a hub and a stiffener within the inner casing;
- (6) FIG. 5 is a cross-sectional view of the stiffener of FIG. 4, illustrating two sealing members positioned around the stiffener;
- (7) FIG. 6 is an enlarged view of a portion of the stiffener of FIG. 5, illustrating two sealing members having different diameters in circumferential grooves having the same depth;
- (8) FIG. 7 is a cross-sectional view of the pipe fitting of FIG. 1, illustrating pipes having a minimum internal diameter capable of fitting over the stiffener;
- (9) FIG. 8 is an enlarged view of a portion of the pipe fitting of FIG. 7, illustrating the two sealing members engaging an inner wall of the pipe having the minimum internal diameter;
- (10) FIG. 9 is a cross-sectional view of the pipe fitting of FIG. 1, illustrating pipes having a maximum internal diameter capable of fitting over the stiffener;
- (11) FIG. 10 is an enlarged view of a portion of the pipe fitting of FIG. 9, illustrating the two sealing members engaging an inner wall of the pipes having the maximum internal diameter;
- (12) FIG. 11 is a cross-sectional view of another exemplary embodiment of the stiffener of FIG. 4, illustrating three sealing members positioned around the stiffener;
- (13) FIG. 12 is an enlarged view of a portion of the stiffener of FIG. 11, illustrating three sealing members having different diameters in circumferential grooves having the same depth;
- (14) FIG. 13 is an enlarged view of a portion of the pipe fitting of FIG. 7 having a stiffener of FIG. 11, illustrating the three sealing members engaging an inner wall of the pipe having the minimum internal diameter;
- (15) FIG. 14 is an enlarged view of a portion of the pipe fitting of FIG. 9 having a stiffener of FIG. 11, illustrating the three sealing members engaging an inner wall of the pipes having the maximum internal diameter;
- (16) FIG. 15 is a cross-sectional view of another exemplary embodiment of a stiffener according to the present disclosure, illustrating two sealing members positioned around the stiffener;
- (17) FIG. 16 is an enlarged view of a portion of the stiffener of FIG. 15, illustrating sealing members have the same diameter in circumferential grooves having different depths;
- (18) FIG. 17 is a cross-sectional view of the pipe fitting of FIG. 1 and the stiffener of FIG. 15, illustrating pipes having a minimum internal diameter capable of fitting over the stiffener;
- (19) FIG. 18 is an enlarged view of a portion of the pipe fitting of FIG. 17, illustrating the two sealing members engaging an inner wall of the pipes having the minimum internal diameter;
- (20) FIG. 19 is a cross-sectional view of the pipe fitting of FIG. 1 and the stiffener of FIG. 15, illustrating pipes having a maximum internal diameter capable of fitting over the stiffener;
- (21) FIG. 20 is an enlarged view of a portion of the pipe fitting of FIG. 19, illustrating the two sealing members engaging an inner wall of the pipes having the maximum internal diameter;

- (22) FIG. 21 is a cross-sectional view of another exemplary embodiment of the stiffener of FIG. 4, illustrating three sealing members positioned around the stiffener;
- (23) FIG. 22 is an enlarged view of a portion of the stiffener of FIG. 21, illustrating sealing members have the same diameter in circumferential grooves having different depths;
- (24) FIG. 23 is an enlarged view of a portion of the pipe fitting of FIG. 7 having a stiffener of FIG. 21, illustrating the three sealing members engaging an inner wall of the pipe having the minimum internal diameter;
- (25) FIG. 24 is an enlarged view of a portion of the pipe fitting of FIG. 9 having a stiffener of FIG. 21, illustrating the three sealing members engaging an inner wall of the pipes having the maximum internal diameter;
- (26) FIG. 25 is a perspective view of another exemplary embodiment of a pipe fitting according to the present disclosure;
- (27) FIG. 26 is a cross-sectional view of the pipe fitting of FIG. 25 taken along line 26-26;
- (28) FIG. 27 is a front perspective view of another exemplary pipe fitting according to the present disclosure;
- (29) FIG. 28 is a cross-sectional view of the pipe fitting of FIG. 27 taken along line 28-28;
- (30) FIG. 29 is a front perspective view of another exemplary pipe fitting according to the present disclosure;
- (31) FIG. 30 is a side view of the pipe fitting of FIG. 29;
- (32) FIG. 31 is a front perspective view of another exemplary pipe fitting according to the present disclosure; and
- (33) FIG. 32 is a side view of the pipe fitting of FIG. 31.

DETAILED DESCRIPTION

(34) The present disclosure provides various embodiments of improvements to stab-type pipe fittings that are capable of being inserted onto plastic pipes and provide a leak proof connection. Exemplary types of the stab-type pipe fittings used to connect to plastic pipe include, but are not limited to, couplers, elbows (also known as “L's”), Tee's, reducers and end caps. Those skilled in the art would readily recognize that the features of the improvements to stab-type pipe fittings disclosed herein may be incorporated into any stab-type pipe fitting, including those noted above. For ease of description, the present disclosure may refer to stab-type pipe fittings as the pipe fitting in the singular or pipe fittings in the plural.

(35) In the exemplary embodiment shown in FIGS. 1-4, the pipe fitting is a coupler used to join two plastic pipes together. Thus, for this part of the description, the pipe fitting will be referred to as the coupler. The coupler 10 includes an outer casing or shell 20, an inner casing or shell 40, one or more grippers 70, and one or more stiffeners 90. The one or more grippers 70 and one or more stiffeners 90 are encased within the inner casing 40 and the inner casing 40 is encased within the outer casing 20.

(36) The outer casing 20 has a body 22 and one or more pipe ports 24. The body 22 has a hollow interior that receives the inner casing 40 to surround the inner components of the coupler 10, as seen in FIG. 3. The body 22 has one or more openings 26 each configured to permit a plastic pipe to pass into the body 22. In the embodiment shown in FIG. 2, the body 22 has two openings 26. It should be noted that for certain types of pipe fittings the body 22 may have more than two openings, such as Tee type pipe fittings, and for other types of pipe fittings the body 22 may have one opening, such as end cap type pipe fittings. The body 22 is made of a sufficiently rigid material to provide a protective casing for the inner components of the coupler 10.

(37) Each of the one or more pipe ports 24, which may also be referred to as a moisture lip, have a center opening 30 and an elongated port lip 34. The center opening 30 is aligned with an opening 26 in the body 22. In the exemplary embodiment of a coupler as the pipe fitting, there are two pipe ports 24. However, it should be noted that for certain types of the pipe fittings the outer casing 20 may have more than two pipe ports 24, such as in Tee type pipe fittings, and in other types of pipe

fittings the outer casing **20** may have one pipe port, such as end cap type pipe fittings. Each pipe port lip **34** may be integrally formed into the outer casing body **22** such that the openings **26** and **30** are one continuous passageway into the outer casing **20**, or each pipe port lip **34** may be secured to the outer casing body **22** using, for example, adhesives, welds or mechanical fasteners. Where a pipe port **24** is secured to the outer casing body **22**, the center opening **30** is aligned with an outer casing opening **26** to form a passageway into the outer casing **20**. Providing a passageway into the outer casing **20** permits a plastic pipe to pass from a pipe port **24** through the opening **26** into the outer casing **20**, and eventually into the inner casing **40**.

(38) As noted, the pipe port lip **34** is an elongated lip that extends from the outer casing body **22**. As shown in FIG. 1, each elongated lip **34** extends from the outer casing body **22** a distance D1 that may range from about $\frac{1}{8}$ of an inch to about 2 inches. Each pipe port lip **34** is configured to receive a pipe **200**, seen for example in FIG. 2, and the elongated nature of the pipe port lip **34** aligns and straightens the pipe **200** as it enters the coupler **10** prior to the pipe passing over the stiffener **90**. The elongated pipe port lip **34** also ensures that the pipe **200** remains substantially straight within the coupler **10**. By maintaining the straightness of the pipe **200** within the coupler **10**, the pipe port lip **34** also ensures that an even radial compression can be maintained between the sealing members of the stiffener **90** and the inner diameter of the pipe **200** even under bending loads. To provide additional structural integrity to each pipe port **24**, one or more gussets **38** may be positioned around the perimeter of each pipe port lip **34**. The gussets **38** may be integrally formed into the pipe port lip **34** and the outer casing body **22**, or the gussets **38** may be secured to the pipe port lip **34** and the outer casing body **22** using, for example, adhesives, welds or mechanical fasteners. To facilitate the described pipe straightening, the pipe port lip **34** and gussets **38**, if used, are preferably made of a sufficiently rigid material.

(39) Referring to FIGS. 2 and 4, as noted, the inner casing **40** of the coupler **10** encases the gripper **70** and at least a portion of the stiffener **90**. The inner casing **40** is a hollow casing or shell that may be formed as a single casing or as a multi-component casing coupled together. The type of pipe fitting may determine whether the inner casing **40** is a single casing or a multi-component casing. For example, the inner casing **40** may be a multi-component casing where the pipe fitting has two or more pipe ports **24**, such as in coupler type pipe fittings, elbow type pipe fittings, Tee type pipe fittings, or reducer type pipe fittings. As another example, where the pipe fitting has one pipe port **24**, such as in end cap type pipe fittings, the inner casing **40** may be a single casing. In the embodiment of a coupler shown in FIGS. 1-4, the inner casing **40** is a multi-component casing having two casing members **42** that are coupled within the outer casing **20** to form the inner casing **40**. In the exemplary embodiment of an elbow shown in FIGS. 19 and 20, the inner casing **40** is a multi-component casing having two casing members **42** and a main body between the casing members to effect a 90-degree angle of the elbow shown. Thus, in this exemplary embodiment the casing members **42** and the main body form the inner casing **40**.

(40) Continuing to refer to FIGS. 2-4, each casing member **42** has an inner end **46** and an outer end **48**. The inner end **46** has a circumferential ring **50** extending beyond an outer surface of the casing member **42**. The ring **50** is aligned with the opening **26** in the outer casing body **22** so that a pipe **200** passing through a pipe port **24** and an opening **26** can pass into the hollow portion of the casing member **42**. On the inside of the circumferential ring **50** is a circumferential notch **52**. In the embodiment of FIGS. 1-4, the circumferential notch **52** is used to receive a hub **60** that maintains the gripper **70** and the stiffener **90** within the inner casing **40**. In the exemplary embodiment of the coupler **10**, the notches **52** of each casing member **42** when joined form a channel **44** to receive an outer wall **62** of the hub **60**. The hub **60** may be secured within the channel **44** by, for example, a friction fit, adhesive, weld or mechanical fastener. The hollow portion of the inner casing members **42** extends from the circumferential notch **52** to a circumferential ring **54** at the outer end **48**. The hollow portion of the casing members **42** has a straight portion starting at the notch **52** followed by a tapered portion, e.g., a frusto-conical surface, that ends at the circumferential ring **54**. The

circumferential ring **54** maintains the gripper **70** within the inner casing **40**. The outer end **48** includes an opening **56** that can receive the stiffener **90** and a pipe **200**.

(41) It is noted that for the coupler **10**, two grippers **70** are incorporated into the coupler **10**, one in each casing member **42**. Each gripper **70** has an opening **72** that is aligned with a pipe port **24**, and includes reverse serrations that can flex when a pipe **200** is passed into the gripper **70** imparting little resistance to the forward advancement of the pipe **200** into the inner casing member **42**, while imparting sufficient resistance to rearward movement of the pipe to prevent withdrawal of the pipe **200** from the inner casing **40**. In instances where a pipe fitting has more than two pipe ports **24**, e.g., a Tee type pipe fitting, there would be more than two grippers **70** within the inner casing **40**, with each gripper **70** opening aligned with a pipe port **24**. In instances where a pipe fitting has one pipe port **24**, e.g., an end cap type pipe fitting, there would be one gripper **70** having an opening **30** that is associated with the pipe port **24**.

(42) The gripper **70** is a hollow member having an inner end **70a** that is configured to abut hub **60** and an outer end **70b**. The side of the gripper **70** opposite the hub **60** is tapered, generally a frusto-conical shape that coincides with the shape and slope of a tapered edge within the casing member **42** adjacent the ring **54**. Each gripper **70** is preferably constructed of material that is harder than the pipe. Having a tapered end of the gripper **70** that is shaped to coincide with the tapered edge of the casing member **42** allows the gripper **70** to engage the casing member **42** in the event a pipe **200** is pulled in an attempt to withdraw the pipe **200** from the coupler **10**. When the gripper **70** tapered end engages the tapered edge of the casing member **42**, the gripper **70** is caused to constrict so that the serrations more tightly bite into the pipe **200** to prevent the withdrawal of the pipe from the coupler **10**.

(43) Referring to FIGS. **5** and **6**, the stiffener **90** has two sets of two sealing members **94** and **96** positioned along the stiffener **90**. However, it should be understood that each set may have more than two sealing members **94** and **96** positioned along the stiffener **90**. For example, FIGS. **11** and **12** show a stiffener **90** having two sets of three sealing members **94**, **95** and **96** positioned along the stiffener **90**. The stiffener **90** is preferably a tubular structure constructed of rigid material, such as plastic or metal. The stiffener **90** is positioned within the inner casing **40** so that one end **98** is positioned in alignment with the opening **26** in the outer casing **20** and the opening **56** in one inner casing member **42**. The other end **100** of the stiffener **90** is positioned in alignment with another opening **26** in the outer casing **20** and opening **56** in the other inner casing member **42**. The stiffener **90** has an external diameter that is smaller than the allowable variation in the internal diameter $D2'$ (seen in FIG. **7**) and $D2''$ (seen in FIG. **9**) of the pipes **200** that may be inserted into the coupler **10** and pushed onto the stiffener **90**. As pipes **200** are inserted into a pipe port **24**, the corresponding opening **26** in the outer casing **20** and the corresponding opening **56** in the other inner casing member **42**, they can be pushed onto the ends **98** or **100** of the stiffener **90**. As noted above with reference to FIG. **2**, as a pipe **200** is pushed into a pipe port **24**, the pipe port **24** straightens the pipe before the pipe is passed over the stiffener **90**.

(44) As described above, the stiffener **90** is secured within coupler **10** using the hub **60** that encircles the stiffener **90** approximately midway between ends **98** and **100**. To assist in securing the hub **60** to the stiffener **90**, a knurled area **102** may be formed on the outer surface of the stiffener **90**, as shown in FIG. **5**. The knurled area **102** helps maintain the hub **60** in position on the stiffener **90**. In another embodiment, the hub **60** may be bonded to the stiffener **90** at the knurled area **102** or instead of including a knurled area.

(45) To seal plastic pipes **200** within the coupler **10** using the stiffener **90** of FIGS. **5** and **6**, the stiffener **90** includes two sets of two circumferential grooves **104** and **106** to support the sealing members **94** and **96**, which may be, for example, elastomeric O-rings. The sealing members **94** and **96** provide a multi-compression feature where both sealing members contact the internal walls of the plastic pipe **200** connected to the coupler **10** to provide a liquid or gas proof seal. In this exemplary embodiment, each pair of circumferential grooves **104** and **106** and corresponding

sealing members **94** and **96** are positioned on the stiffener **90** on opposite sides of the location where the hub **60** would be located relative to the stiffener **90**, or in other words on opposite sides of the knurled area **102** if the stiffener **90** has a knurled area **102**. The circumferential grooves **104** and **106** are formed in an outer surface of the stiffener **90**, as shown in FIGS. 5 and 6. Each pair of circumferential grooves **104** and **106** may also be referred to as an inner circumferential groove **104** and an outer circumferential groove **106**. The circumferential grooves **104** and **106** are spaced apart a distance “D3” that may range from about 5 mm to about 50 mm. To seal plastic pipes **200** within the coupler **10** using the stiffener **90** of FIGS. 11 and 12, the stiffener includes two sets of three circumferential grooves **104**, **105** and **106** to support the sealing members **94**, **95** and **96**, which may be, for example, elastomeric O-rings. The sealing members **94**, **95** and **96** provide a multi-compression feature where all three sealing members may contact the internal walls of the plastic pipe **200** connected to the coupler **10** to provide a liquid or gas proof seal. In this exemplary embodiment, each pair of circumferential grooves **104**, **105** and **106** and corresponding sealing members **94**, **95** and **96** are positioned on the stiffener **90** on opposite sides of the location where the hub **60** would be located relative to the stiffener **90**, or in other words on opposite sides of the knurled area **102** if the stiffener **90** has a knurled area **102**. The circumferential grooves **104**, **105** and **106** are formed in an outer surface of the stiffener **90**, as shown in FIGS. 11 and 12. Each pair of circumferential grooves may also be referred to as an inner circumferential groove **104**, an intermediate circumferential groove **106**, and an outer circumferential groove **105**. The circumferential grooves **104**, **105** and **106** are spaced apart a distance “D3” that may range from about 5 mm to about 50 mm.

(46) The multi-compression feature of the present disclosure permits the pipe fitting, e.g., the coupler **10**, to connect to pipes **200** with variations in the internal diameter of the pipe and maintain a sufficient seal to prevent liquid or gas leakage. For example, one vendor could make a 1” polyethylene pipe with an internal diameter of 1.082 inches, and another vendor could make a 1” polyethylene pipe with an internal diameter 1.044 inches. Both pipes **200** would be within the dimensional requirements for such 1” pipes, but that variation in the internal diameters may be sufficient to impact the seal between the pipe and the fitting. As another example, when a 1” polyethylene pipe is manufactured the 1” pipe may have variations in the inner diameter of the pipe due to irregularities in the manufacturing process.

(47) In addition, the multi-compression feature of the present disclosure permits the pipe fitting, e.g., the coupler **10**, to maintain the seal between the fitting and the pipe **200** in an expanded temperature range. For example, in one temperature range, e.g., -40 degrees F. to 100 degrees F., a sealing member **94**, **95** or **96**, such as an elastomeric O-ring, may retain its hardness, elongation and compressive properties, but at a different temperature range, e.g., 100 degrees F. to 200 degrees F., the sealing member's hardness, elongation and compressive properties may change. Using the multi-compression feature of the present disclosure allows the use of different size sealing members **94**, **95** or **96** or positions the sealing members differently to meet potential variations in the sealing members that may be caused by temperature fluctuations.

(48) To be able to seal pipes **200** with variations in their internal diameter and/or in expanded temperature range scenarios, the stiffener **90** shown in FIGS. 5 and 6, for example, may have circumferential grooves **104** and **106** that are the same size while the sealing members **94**, **95** or **96**, e.g., elastomeric O-rings, are different in sizes. To illustrate, circumferential grooves **104** and **106** have a width W1 that may range from about 1 mm to about 20 mm, and a height H1 that may range from about 1 mm to about 20 mm. The sealing members **94** and **96** are different in that the outer sealing member **96** is smaller than the inner sealing member **94** to facilitate easier insertion of a pipe **200** onto the stiffener **90**. In this embodiment and referring to FIGS. 7 and 8, a pipe **200** having a minimum internal diameter variation D2’ that may fit over the stiffener **90** causes the sealing members **94** and **96** to compress to a maximum compression to seal the pipe **200** relative to the stiffener **90**. In this example, the sealing member **94** compresses more than sealing member **96**,

as shown. Similarly, and referring to FIGS. 9 and 10, a pipe 200 having a maximum internal diameter variation $D2''$ that may fit over the stiffener 90 causes the sealing members 94 and 96 to compress to a minimum compression to seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 94 compresses slightly more than sealing member 96, as seen in FIGS. 9 and 10.

(49) To be able to seal pipes 200 with variations in their internal diameter and/or in expanded temperature range scenarios, the stiffener 90 shown in FIGS. 11 and 12, for example, may have circumferential grooves 104, 105 and 106 that are the same size while the sealing members 94, 95 or 96, e.g., elastomeric O-rings, are different in sizes. To illustrate, circumferential grooves 104, 105 and 106 have a width $W1$ that may range from about 1 mm to about 20 mm, and a height $H1$ that may range from about 1 mm to about 20 mm. The sealing members 94, 95 and 96 are different in that the outer sealing member 95 is smaller than the intermediate sealing member 96, which is smaller than the inner sealing member 94 to facilitate easier insertion of a pipe 200 onto the stiffener 90. In this embodiment and referring to FIG. 13, a pipe 200 having a minimum internal diameter variation $D2'$ that may fit over the stiffener 90 causes the sealing members 94, 95 and 96 to compress to a maximum compression to seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 96 compresses more than sealing member 95, and the sealing member 94 compresses more than sealing member 96, as shown. Similarly, and referring to FIG. 14, a pipe 200 having a maximum internal diameter variation $D2''$ that may fit over the stiffener 90 causes the sealing members 94 and 96 to compress to a minimum compression while sealing member 95 does not engage the pipe 200 so that sealing members 94 and 96 seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 94 compresses slightly more than sealing member 96, as seen in FIG. 14.

(50) Referring now to FIGS. 4, 15 and 16, another exemplary embodiment of a stiffener 90 for sealing pipes 200 with variations in their internal diameter and/or in expanded temperature range scenarios is shown. In this embodiment, the stiffener 90 has two sets of two circumferential grooves 104 and 106 that differ in size but receive sealing members 94 or 96 having the same size, e.g., the same diameter. To illustrate, circumferential grooves 104 and 106 have a width $W2$ that may range, for example, from about 1 mm to about 10 mm, but circumferential groove 104 has a height $H2$ that may range from about 0.6 mm to about 10 mm, while circumferential groove 106 has a height $H3$ that may range from about 0.9 mm to about 11 mm. The two sets of sealing members 94 and 96, e.g., elastomeric O-rings, are the same diameter size, e.g., the same diameter. As a result, and referring to FIGS. 17 and 18, a pipe 200 having a minimum internal diameter variation $D2'$ that may fit over the stiffener 90 causes the sealing members 94 and 96 to compress to a maximum compression to seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 94 compresses more than sealing member 96. Similarly, and referring to FIGS. 19 and 20, a pipe 200 having a maximum internal diameter variation $D2''$ that may fit over the stiffener 90 causes the sealing members 94 and 96 to compress to a minimum compression for this embodiment to seal the pipe 200 to the stiffener 90. In this example, the sealing member 94 compresses more than sealing member 96.

(51) Referring now to FIGS. 21 and 22, another exemplary embodiment of a stiffener 90 for sealing pipes 200 with variations in their internal diameter and/or in expanded temperature range scenarios is shown. In this embodiment, the stiffener 90 has two sets of three circumferential grooves 104, 105 and 106 that differ in size but receive sealing members having the same size, e.g., the same diameter. To illustrate, circumferential grooves 104, 105 and 106 have a width $W2$ that may range from about 1 mm to about 10 mm, but circumferential groove 104 has a height $H2$ that may range from about 0.6 mm to about 10 mm, while circumferential groove 106 has a height $H3$ that may range from about 0.9 mm to about 11 mm, while circumferential groove 105 has a height $H4$ that may range from about 1.2 mm to about 12 mm. The two sets of sealing members 94, 95 and 96, e.g., elastomeric O-rings, are the same diameter size, e.g., the same diameter. As a result, and

referring to FIG. 23, a pipe 200 having a minimum internal diameter variation D2' that may fit over the stiffener 90 causes the sealing members 94, 95 and 96 to compress to a maximum compression to seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 94 compresses more than sealing member 96, and the sealing member 96 compresses more than sealing member 95. Similarly, and referring to FIG. 24, a pipe 200 having a maximum internal diameter variation D2'' that may fit over the stiffener 90 causes the sealing members 94 and 96 to compress to a minimum compression while sealing member 95 does not engage the pipe 200 so that sealing members 94 and 96 seal the pipe 200 relative to the stiffener 90. In this example, the sealing member 94 compresses more than sealing member 96.

(52) While the stiffener and sealing members that provide the multi-compression feature of the present disclosure are shown as having two or three sealing members on each side of the stiffener, the present disclosure contemplates using more than three sealing members on each side of the stiffener.

(53) Turning to FIGS. 25 and 26, another exemplary embodiment of a pipe fitting according to the present disclosure is shown. In the exemplary embodiment shown, the pipe fitting is a coupler 120 that is used to join two plastic pipes 200 together. Thus, for this part of the description, the pipe fitting will be referred to as the coupler 120. The coupler 120 includes an outer casing or shell 20, an inner casing or shell 40, a gripper 70, and a stiffener 130. In this exemplary embodiment, the outer casing 20 is substantially similar to the outer casing 20 described above, except that the pipe port lips 34 are not an elongated body. In this exemplary embodiment, the pipe ports lips 34 are more like snub-nose bodies similar to those in conventional stab-type pipe fittings. In addition, the inner casing 40 and gripper 70 are the same as the inner casing and gripper described above and for ease of description are not repeated. In this embodiment, the stiffener 130 is an elongated stiffener that extends outside the outer casing 20 a distance D4 which may be at least about 1 inch. The elongated stiffener 130 ensures that a pipe 200 connected to the coupler 120 is straight at the point where the sealing members 94 and 96 engage the inner wall of the pipe 200. To seal the pipe 200 to the stiffener 130, the embodiments of the circumferential grooves 104 and 106 and the sealing members 94 and 96 described above may be used and are not repeated. However, as noted above, while the stiffener 130 and sealing members 94 and 96 that provide the multi-compression feature of the present disclosure are shown in FIGS. 25 and 26 as having two sealing members 94 and 96 on each side of the stiffener 130, the present disclosure contemplates using more than two sealing members on each side of the stiffener 130, such as three sealing members on each side of the stiffener 130.

(54) Referring now to FIGS. 27 and 28, another exemplary embodiment of a pipe fitting according to the present disclosure is shown. In this exemplary embodiment, the pipe fitting is an elbow 140 used to join two plastic pipes together at an angle. Examples of elbows include 90 degree elbows, 45 degree elbows and 22 degree elbows. Thus, for this part of the description, the pipe fitting will be referred to as the elbow 140 where like numbered components have like functions as described above with reference to FIGS. 1-26. The outer casing 20 in this exemplary embodiment is shaped as an elbow with the desired angle, here 90 degrees. The elbow 140 in this embodiment has two pipe ports 24. To accommodate the bend of the elbow, the casing members 42 of the inner casing 40 are separated and a main body 142 having fluid or gas passageway 144 is inserted between the casing members 42 which together form the inner casing 40. Retained within each casing member 42 is a gripper 70 having the components and functions as described above.

(55) In this exemplary embodiment, two stiffeners 150 are provided, where one stiffener 150 is associated with one casing member 42 and the other stiffener 150 is associated with the other casing member 42. As noted, to connect one stiffener 150 to the other stiffener 150 the inner casing 40 includes the main body 142 having a fluid or gas passageway 144 that provides a conduit in the shape of the angle of the elbow 140, e.g., 90 degrees, to permit fluid or gas to pass from one stiffener 150 to the other stiffener 150, and thus from one pipe 200 to another pipe 200. Each

stiffener **150** is substantially similar to a portion of the stiffener **90** described above. More specifically, each stiffener **150** is a tubular member made of a rigid material, and includes at least two sealing members, e.g., sealing members **94** and **96**, supported within at least two corresponding circumferential grooves, e.g., grooves **104** and **106**, as described above. However, as noted above, while the stiffener **150** and sealing members **94** and **96** that provide the multi-compression feature of the present disclosure are shown in FIGS. **27** and **28** as having two sealing members **94** and **96** on each side of the stiffener **150**, the present disclosure contemplates using more than two sealing members on each side of the stiffener **150**, such as three sealing members on each side of the stiffener **150**. To secure the stiffener **150** within the inner casing **40**, an end **154** of the stiffener **150** may include a knurled area **156** that secures the stiffener **150** within a hollow portion of the main body **142** by, for example, a friction fit as shown in FIG. **21**. In another embodiment, the stiffener **150** may be secured within the hollow portion of the main body **142** using an adhesive.

(56) Referring now to FIGS. **29** and **30**, another exemplary embodiment of a pipe fitting according to the present disclosure is shown. In this exemplary embodiment, the pipe fitting is a Tee used to join three plastic pipes together. Thus, for this part of the description, the pipe fitting will be referred to as the Tee **160** where like numbered components have like functions as described above with reference to FIGS. **1-20**. The outer casing **20** in this exemplary embodiment is in the shape of a “T”. In this embodiment, there are three casing members **42** to accommodate the bends of the Tee. The casing members **42** are separated by a main body **142** having fluid or gas passageway inserted between the casing members **42** similar to the embodiment described with reference to the elbow **140** of FIGS. **25** and **26**, but adopted for a Tee. Together the casing members **42** and main body **142** form the inner casing **40**. Retained within each casing member **42** is a gripper having the function as the gripper **70** described above.

(57) In this exemplary embodiment, three stiffeners **150** similar to the stiffeners **150** described above are provided, where one stiffener **150** is associated with one casing member **42**. As noted, to connect one stiffener **150** to the other stiffeners the inner casing **40** includes the main body **142** that provides a conduit in the shape of a Tee to permit fluid or gas to pass from one stiffener **150** to the other stiffeners **150**. As noted, each stiffener is substantially similar to the stiffener **150** described above with reference to FIG. **26**. More specifically, each stiffener **150** is a tubular member made of a rigid material, and includes at least two sealing members, e.g. sealing members **94** and **96**, supported within at least two corresponding circumferential grooves, e.g., grooves **104** and **106**, as described above. However, as noted above, while the stiffener **150** and sealing members **94** and **96** that provide the multi-compression feature of the present disclosure are shown in FIGS. **29** and **30** as having two sealing members on each side of the stiffener **150**, the present disclosure contemplates using more than two sealing members on each side of the stiffener **150**, such as three sealing members on each side of the stiffener. To secure the stiffener **150** within the inner casing **40**, an end **154** of the stiffener **150** may include a knurled area **156** that secures the stiffener **150** within a hollow portion of the main body by a friction fit. In another embodiment, the stiffener **150** may be secured within the hollow portion of the main body using an adhesive.

(58) Referring now to FIGS. **31** and **32**, another exemplary embodiment of a pipe fitting according to the present disclosure is shown. In this exemplary embodiment, the pipe fitting is an end cap **180** used to terminate a plastic pipe. Thus, for this part of the description, the pipe fitting will be referred to as the end cap **180** where like numbered components have like functions as described above with reference to FIGS. **1-30**. The outer casing **20** in this exemplary embodiment has a closed side **23** and a side with a pipe port **24**. In this exemplary embodiment, there is one casing member forming the inner casing **40** within the outer casing **20**. Encased within the inner casing **40** is a gripper similar to gripper **70** described above and shown in FIG. **4**, and a stiffener **150** similar to stiffener **150** described above and shown in FIG. **26**. More specifically, the stiffener is a tubular member made of a rigid material, and includes at least two sealing members, e.g., sealing members **94** and **96**, supported within at least two corresponding circumferential grooves, e.g., grooves **104**

and **106**, as described above. However, as noted above, while the stiffener **150** and sealing members **94** and **96** that provide the multi-compression feature of the present disclosure are shown in FIGS. **31** and **32** as having two sealing members on each side of the stiffener, the present disclosure contemplates using more than two sealing members on each side of the stiffener, such as three sealing members on each side of the stiffener. To secure the stiffener **150** within the inner casing **40**, an end **154** of the stiffener **150** may include a knurled area **156** that secures the stiffener **150** within a hollow portion of the casing member **42** by a friction fit. In another embodiment, the stiffener **150** may be secured within the hollow portion of the casing member **42** using an adhesive. (59) It is noted that a more detailed description of the inner casing and gripper, as well as portions of the outer casing and stiffener described herein is are described in commonly owned U.S. Pat. No. 5,692,785 which is incorporated herein in its entirety by reference.

(60) As shown throughout the drawings, like reference numerals designate like or corresponding parts. While illustrative embodiments of the present disclosure have been described and illustrated above, it should be understood that these are exemplary of the disclosure and are not to be considered as limiting. Additions, deletions, substitutions, and other modifications can be made without departing from the spirit or scope of the present disclosure. Accordingly, the present disclosure is not to be considered as limited by the foregoing description.

Claims

1. A pipe fitting comprising: an outer casing having a first port opening and a second port opening; a hollow inner casing positioned within the outer casing and having a first opening and a second opening; first and second grippers positioned within the inner casing, wherein the first gripper has an opening aligned with the first port opening, and the second gripper has an opening aligned with the second port opening; and a stiffener having a first end and a second end, the stiffener being positioned within the first and second gripper openings such that the stiffener is aligned with the first and second port openings, the stiffener including: a first pair of grooves in an outer surface of the first end of the stiffener and a first pair of sealing members, wherein one sealing member in the first pair of sealing members is positioned within one of the grooves of the first pair of grooves; and a second pair of grooves in an outer surface of the second end of the stiffener and a second pair of sealing members, wherein one sealing member in the second pair of sealing members is positioned within one of the grooves of the second pair of grooves; wherein the first pair of grooves are positioned on the stiffener so that when a first pipe is inserted into the first port opening, the first pipe passes over the first end of the stiffener and each sealing member in the first pair of sealing members and into the first gripper so that each sealing member in the first pair of sealing members contacts an interior wall of the first pipe, and wherein the second pair of grooves are positioned on the stiffener so that when a second pipe is inserted into the second port opening, the second pipe passes over the second end of the stiffener and each sealing member in the second pair of sealing members and into the second gripper so that each sealing member in the second pair of sealing members contacts an interior wall of the second pipe.
2. The pipe fitting according to claim 1, wherein each sealing member in the first pair of sealing members comprises an elastomeric O-ring.
3. The pipe fitting according to claim 1, wherein each sealing member in the second pair of sealing members comprises an elastomeric O-ring.
4. The pipe fitting according to claim 1, further comprising a hub positioned on the stiffener so that the hub is between the first gripper and the second gripper, the hub being configured to maintain the first and second grippers and the stiffener within the inner casing.
5. The pipe fitting according to claim 1, wherein the first port opening is at a right angle relative to the second port opening.
6. The pipe fitting according to claim 1, wherein the pipe fitting comprises a coupler.

7. The pipe fitting according to claim 1, wherein each sealing member in the first pair of sealing members have the same diameter.
 8. The pipe fitting according to claim 1, wherein each sealing member in the second pair of sealing members have the same diameter.
 9. A pipe fitting comprising: an outer casing having at least one elongated lip that includes a port opening, the at least one elongated lip being sufficient in length to align and straighten a pipe as it enters the port opening; a hollow inner casing positioned within the outer casing and having at least one opening aligned with the port opening in the at least one elongated lip; at least one gripper positioned within the inner casing and having an opening aligned with the port opening in the at least one elongated lip, the at least one gripper being configured to grip the pipe inserted into the at least one gripper opening to prevent withdrawal of the pipe from the at least one gripper; and a stiffener having at least one pair of grooves in an outer surface of the stiffener and at least one pair of sealing members, wherein one sealing member in the at least one pair of sealing members is positioned within one of the grooves of the at least one pair of grooves, the stiffener being positioned within the opening of the at least one gripper and aligned with the port opening so that a pipe inserted into the port opening can pass into the at least one gripper and over an end of the stiffener having the at least one pair of grooves and the at least one pair of sealing members within the at least one gripper such that the at least one pair of sealing members can contact an interior wall of the pipe to provide a seal between the pipe and the stiffener.
 10. The pipe fitting according to claim 9, wherein the at least one elongated lip further comprises at least one gusset positioned around the port opening of the at least one elongated lip.
 11. The pipe fitting according to claim 9, wherein the at least a pair of sealing members comprise elastomeric O-rings.
 12. The pipe fitting according to claim 9, wherein the pipe fitting comprises a coupler.
 13. A pipe fitting comprising: an outer casing having at least one elongated lip with a port opening forming at least part of at least one passageway, the elongated lip being sufficient in length to align and straighten a pipe as it enters the passageway; a hollow inner casing positioned within the outer casing and having at least one opening aligned with the at least one passageway; at least one gripper positioned within the inner casing and having an opening aligned with the at least one passageway; and a stiffener positioned at least partially within the inner casing and the at least one gripper opening, the stiffener being aligned with the at least one passageway such that when the pipe is inserted into the at least one passageway the pipe passes through the inner casing and over the stiffener and through the at least one gripper, wherein the stiffener has at least one pair of sealing members, each sealing member of the at least one pair of sealing members is positioned within a groove in an outer surface of the stiffener, the grooves being positioned on the stiffener such that when the pipe is passed over the stiffener and sealing members, each sealing member contacts an interior wall of the pipe.
 14. The pipe fitting according to claim 13, wherein the elongated lip further comprises at least one gusset positioned around the port opening.
 15. The pipe fitting according to claim 13, wherein the at least a pair of sealing members comprise elastomeric O-rings.
 16. The pipe fitting according to claim 13, wherein the pipe fitting comprises a coupler.
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